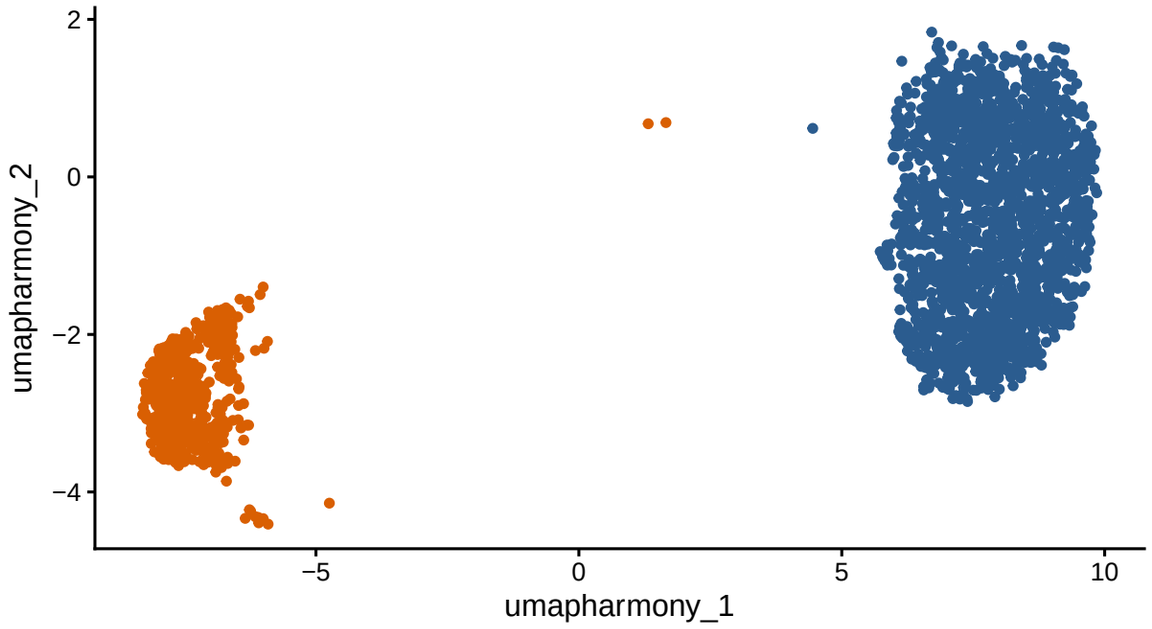
Figure 3: Monocle3 Trajectory Inference of T−Cell Exhaustion in Human TME

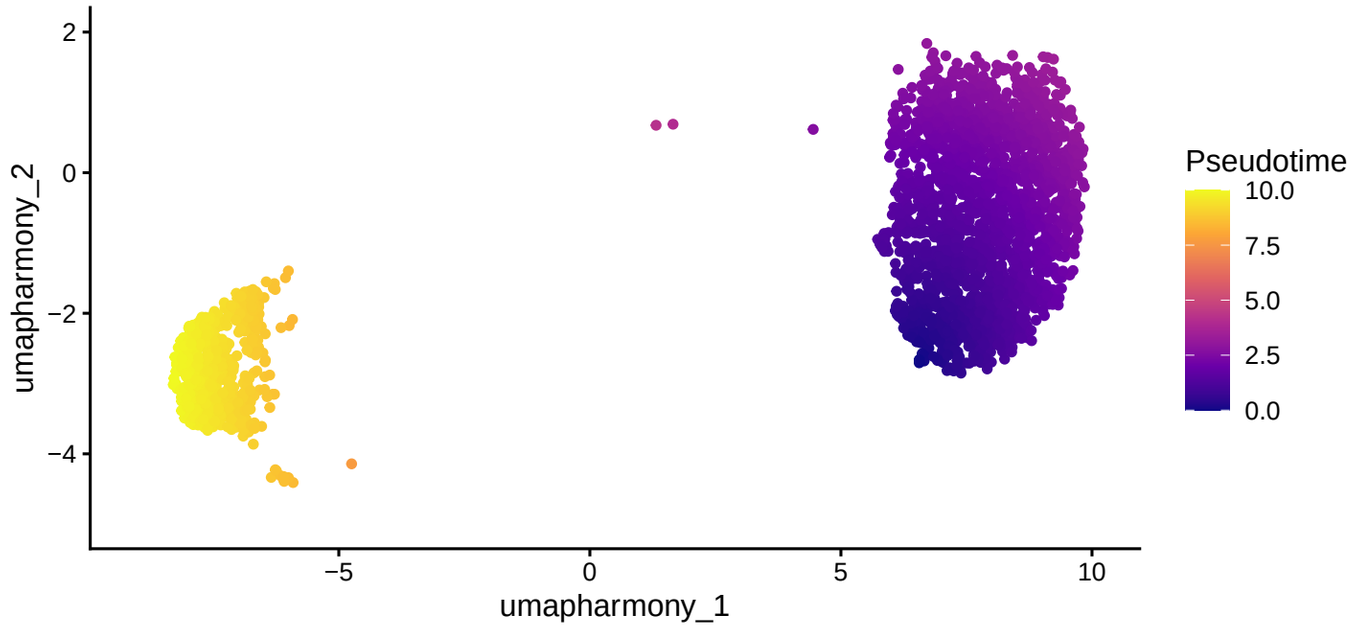
A. CD8+ T−Cell Phenotypic Subsets

Cytotoxic Effectors vs Exhausted TILs



B. Inferred Monocle3 Pseudotime

Geodesic progression (0 = Root / Early Effector −> 10 = Terminal Exhaustion)



CD8+ Cytotoxic T CD8+ Exhausted T (PD−1+)

C. Dynamic Expression Kinetics of Immune Checkpoints vs. Lytic Molecules

GAM cubic spline curves: Progressive upregulation of PD−1, TIM−3 & LAG3 with effector decline

